Lesson 3.8 - Formulas of Statistics in Excel

> **Video:** Statistics - Excel by The Organic Chemistry Tutor **Chapter:** 3 - Central Tendency

The video applies Excel's built-in functions to a column of **19 test scores** in cells `A2:A20`, ranging from 52 to 99. Every function has the same pattern: type `=`, the function name, then select the data range.

Basic summary functions

What you want	Excel formula	Result in the video
Sum	`=SUM(A2:A20)`	—
Number of values n	`=COUNT(A2:A20)`	19
Mean	`=AVERAGE(A2:A20)`, or `=SUM(...)/COUNT(...)`	—
Median	`=MEDIAN(A2:A20)`	84
Mode	`=MODE(A2:A20)` (or `=MODE.SNGL`)	86 (the only score that appears twice)
Minimum	`=MIN(A2:A20)`	52
Maximum	`=MAX(A2:A20)`	99
Range	`=MAX(A2:A20)-MIN(A2:A20)`	47

`MIN` and `MAX` are especially useful with large data sets (thousands of rows) where you can't spot the extremes by eye.

Spread: standard deviation and variance

What you want	Sample (usual choice)	Population
Standard deviation	`=STDEV.S(A2:A20)` → 12.46	`=STDEV.P(A2:A20)`
Variance	`=VAR.S(A2:A20)`	`=VAR.P(A2:A20)`

Variance = SD^2, so if the SD is in cell `E10`, then `=E10^2` gives the same value as `VAR.S`.

Quartiles

`=QUARTILE(A2:A20, k)`, where `k` is 0-4. Newer versions also have `QUARTILE.INC` (same as `QUARTILE`) and `QUARTILE.EXC`, which give slightly different answers.

k	Meaning	Equivalent	Video result
0	0th percentile	minimum	52
1	Q_1, 25th percentile		75
2	Q_2, 50th percentile	median	84
3	Q_3, 75th percentile		88.5
4	100th percentile	maximum	99

Percentiles

`=PERCENTILE(A2:A20, p)` (or `PERCENTILE.INC`). Enter p as a **decimal**: the 25th percentile is `0.25`, **not** `25`.

Percentile	Formula	Same as
0th	`=PERCENTILE(A2:A20, 0)`	minimum, `QUARTILE(...,0)`
25th	`=PERCENTILE(A2:A20, 0.25)`	Q_1
40th	`=PERCENTILE(A2:A20, 0.40)`	
50th	`=PERCENTILE(A2:A20, 0.5)`	median, Q_2
75th	`=PERCENTILE(A2:A20, 0.75)`	Q_3
90th	`=PERCENTILE(A2:A20, 0.9)`	(94.4 in the video)
100th	`=PERCENTILE(A2:A20, 1)`	maximum

Deciles

Deciles split the data into **10** equal parts ("deci" = one tenth), just as quartiles split it into 4. Excel has no decile function, so use `PERCENTILE`:

Decile	Formula
D_0	`=PERCENTILE(A2:A20, 0)` (minimum)
D_1	`=PERCENTILE(A2:A20, 0.1)`

Decile	Formula
D_5	`=PERCENTILE(A2:A20, 0.5)` (median)
D_7	`=PERCENTILE(A2:A20, 0.7)`
D_{10}	`=PERCENTILE(A2:A20, 1)` (maximum)

In general, decile k is `PERCENTILE(range, k/10)`.

Key takeaways

- `SUM`, `COUNT`, `AVERAGE`, `MEDIAN`, `MODE`, `MIN`, `MAX` cover the basics.
- Use `STDEV.S` / `VAR.S` for **samples** and `STDEV.P` / `VAR.P` for **populations**.
- `QUARTILE(range, 0…4)`, `PERCENTILE(range, decimal)`. Deciles = `PERCENTILE(range, k/10)`.
- Excel's quartile and percentile rules can differ slightly from hand methods, because there are several conventions.